Indoor localization using CSI Fingerprint and ML

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***Abstract*—This paper studies indoor Wi-Fi localization based on Channel State Information (CSI) and supervised machine learning. The task is formulated as a multi-class classification problem, with the goal of identifying the position of a device among ten experimental locations. Raw CSI measurements are transformed into compact fingerprints through feature engineering, such as amplitude and phase extraction, normalization, DCT-based denoising, and RSSI-related features. Several classification models are evaluated, including k-Nearest Neighbors, SVM, Random Forests, and XGBoost. Performance is evaluated using macro-averaged F1-score with both K-Fold and Group K-Fold cross-validation to ensure a realistic evaluation. The results show that XGBoost, combined with engineered CSI features, provides robust and stable localization performance.**

I. Introduction

Indoor localization is a challenging problem because GPS signals are not reliable in enclosed environments, while accurate positioning is essential for many real-world applications. Indoor localization has applications in areas such as smart home automation, industrial equipment tracking, and hospital patients monitoring. Wi-Fi-based sensing is attractive for these scenarios, as it leverages existing wireless infrastructure and hardware. The effectiveness of Wi-Fi signal features for machine learning–based localization has been demonstrated in several research works [1].

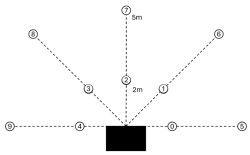

Fig. 1. Experiment to gather CSI data for ten position labels

This report investigates indoor localization as a classification problem using Wi-Fi Channel State Information (CSI), to predict position labels from the experimental data (Fig. 1). CSI provides detailed measurements of the wireless channel, which can be transformed into informative fingerprints that encode spatial properties, including distance and angle.

To model the localization capability, raw CSI is converted into usable generalized features. These features are then used to train machine learning classifiers, within a structured pipeline consisting of preprocessing, feature extraction, cross-validation (K-Fold and Group K-Fold), training, prediction, and evaluation (Fig. 2). This approach serves as a foundation not only for localization, but also for future extensions toward dynamic movement recognition and human tracking.

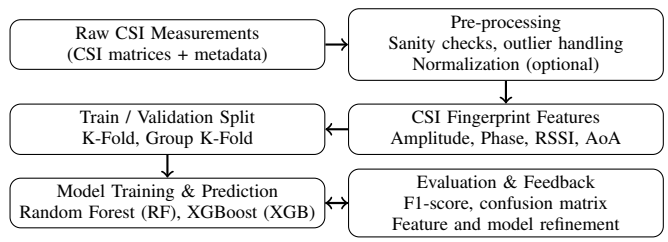

Fig. 2. Overview of the CSI-based localization and classification pipeline.

II. Feature Analysis

1) CSI Amplitude and Phase Extraction: Raw CSI measurements consist of complex-valued in-phase (I) and quadrature (Q) components across multiple antennas and subcarriers, resulting in high dimensional data that are difficult to interpret directly. To obtain more compact representations, the CSI is transformed into amplitude and phase components, preserving key channel characteristics. Packet-level amplitude normalization is applied across subcarriers to capture the shape of the CSI profile rather than its absolute signal strength [2].

Fig. 3 presents a summary of the main derived fields used for feature engineering. These fields and their transforms characterize the wireless channel, providing a compact fingerprint for localization [3].

Feature Extraction	Formula	Terms
Amplitude	$amp_{n,x} = \sqrt{I^2{}_{n,x} + Q^2{}_{n,x}}$	$I = In - Phase\ component\ of\ CSI$ $Q = quadrature\ component\ of\ CSI$
Phase	$phase_{n,x} = arctan2(Q_{n,X}, I_{n,X})$	
AOA	$\Delta\phi_k = \phi_{2,k} - \phi_{1,k}$ $\theta_{n,x} = arcsin(\frac{\lambda}{2\pi d}\Delta\phi)$	$\Delta\phi_k = phase\ diff\ by\ sub\ carriers$ $\theta = AOA;\ \lambda = wavelength$ $d = antenna\ distance$
Phase Slope & Delay	$\frac{d\phi}{dt} = \sim\frac{slope}{\Delta f}$ $\tau = \sim -\frac{1}{2\pi}\frac{slope}{\Delta f}$	$\frac{d\phi}{dt} = rate\ of\ change\ in\ phase$ $\tau = phase\ delay$
Power Spectral Density (PSD)	$PSD = \lvert FFT(normalized\ Amp)\rvert^2$ $max\ (PSD)$	*FFT = Fast Fourier Transform* *how much variation happens at different frequencies along the subcarrier index*

Fig. 3. Derived physical signal characteristics.

2) Amplitude Denoising and Dimensionality Reduction: As observed in the amplitude versus subcarrier plot (Fig. 4), the within-packet amplitude values are often erratic, showing

sharp fluctuations caused by measurement noise. To reduce this instability, the Discrete Cosine Transform (DCT) is applied, which represents each amplitude packet as a sum of smooth, low-frequency cosine components, preserving the structure of the CSI fingerprint while reducing high-frequency spikes. Dimensionality is reduced by keeping the optimal level of cosine components (chosen k=13 as it retains $\tilde{9}9\%$ energy).

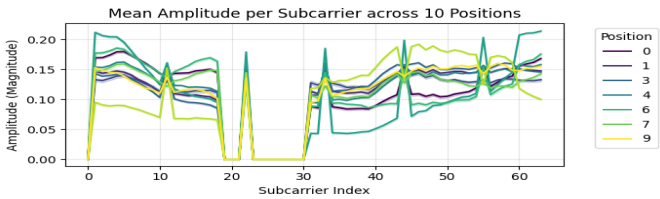

Fig. 4. CSI amplitude versus subcarrier index for one antenna.

The effect of denoising and dimensionality reduction is illustrated using t-SNE visualization (Fig. 5) [4]. The denoised representation demonstrates better structure within the feature space, later demonstrated by increase in f1-score (Tab. IV)

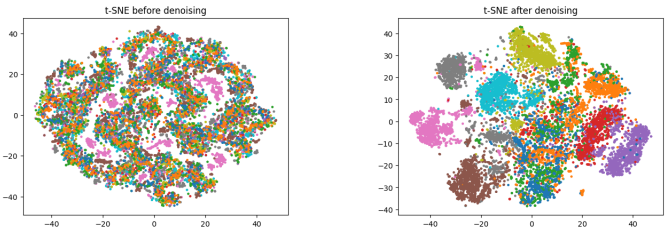

Fig. 5. t-SNE before and after denoising.

3) Assessing RSSI and AOA Discrimination Capacity: The Received Signal Strength Indicator (RSSI) provides coarse spatial information, showing an inverse relationship with the distance from Wi-Fi router across positions (Fig. 6). In contrast, Angle of Arrival (AoA) features show weaker class separability when used alone. To improve angular discrimination, derived fields such as phase slope and phase delay differences between antennas were explored, along with interaction-based features such as AoA × RSSI, which combine direction with signal attenuation to build a more informative data representation.

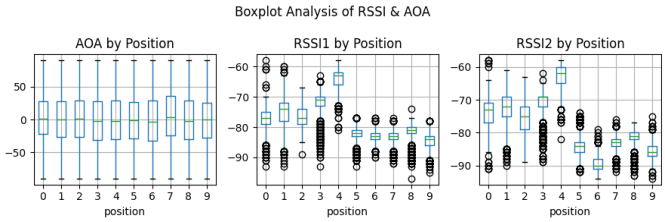

Fig. 6. AoA and RSSI Boxplot by position.

III. Classification Methods

1) Default Baseline models: Multiple classifiers were evaluated to solve the CSI-based localization problem. Simple linear classifiers and distance-based baselines were found to be insufficient since CSI fingerprints create complex and overlapping distributions. For model ensembling, the Bagging (RF) and Boosting (XGB) methods were explored. The baseline model scores (Tab. I) illustrate the need for models that handle non-linear decision boundaries.

TABLE I
Baseline Macro F1-score using K-Fold cross-validation.

Classifier	Macro F1-score
KNN (k=5)	0.838
SVM (RBF)	0.905
Random Forest (RF)	0.865
XGBoost (XGB)	**0.969**

2) Bagging and Support Vector Machines (SVM): RF is a bagging method that trains multiple decision trees in parallel using random data subsets. This reduces variance and makes the model more stable. However, RF still overfits on high-dimensional and correlated data. In addition, similar local structures in the feature space can be inferred from the T-sne (Fig. 5), which possibly gives insight into relatively better SVM performance.

3) Adopting Boosting as the Suitable Model: Boosting (XGBoost) trains trees sequentially, focusing on correcting the errors made by previous trees, progressively reducing bias. XGB, in the baseline analysis, provides the best macro-f1. Although SVM performs well, XGB offers better scalability to larger datasets, flexibility in hyperparameter tuning, and strong interpretability using algorithms such as SHAP. Hence, XGB was adopted as the primary classification method.

4) Handling Class Imbalance: As shown in Tab. II, the data set is moderately balanced between classes. To specifically account for relative imbalance (Ex: position 0 v/s 1), the `class_weight` parameter was set to `"balanced"` within XGB. This improved recall capacity by internally assigning a higher penalty to misclassified minority classes.

TABLE II
Summary statistics of the number of samples per position.

Sample sizes across positions	Value
Location with smallest sample size (pos 0)	1200
Location with largest sample size (pos 1)	1488
Mean sample size	1288.8
Std. Dev. of sample sizes	89.5

IV. Evaluation Metrics

1) Macro-F1: F1 is the harmonic mean of precision and recall. Unlike accuracy, F1 gives equal importance to every position in the localization problem. For this reason, it was chosen as the main metric that guided the cross-validation for both feature selection and model adoption.

2) Model Loss Function: While Macro F1 is used for model comparison, XGBoost optimizes loss functions such as Logloss (binary log loss) and Multiclass logloss (mlogloss). Since CSI localization is a multi-class classification, mlogloss is used as it exponentially penalizes incorrect predictions, producing well-calibrated probabilities.

3) Confusion Matrix (CM): To understand the class-by-class breakdown of misclassifications and highlight systematic errors, confusion matrices were used as a diagnostic tool. Confusions were largely observed between nearby positions with similar distances and angular characteristics (Ex: pairs of locations such as (0,1), (9,4) from Fig. 1).

4) Receiver Operator Characteristic (ROC: one v/s all): Provides the true-positivity vs False-positivity rate for each position labels. The three lowest performing classes from cross validation were flagged for CM analysis. (Ex: Position 0 underperformed across folds).

5) SHapley Additive exPlanations(SHAP) Analysis: SHAP identifies which features strongly influence the model output, enabling Class-wise feature inspection (Ex: what separates class 0 vs class 1), or position targeted feature engineering. For instance, features such as ratios on specific coefficients of low-frequency amplitudes were added to capture subcarrier-shape for class 0. For Improvements from this addition refer heat map (Fig. 7). Additional new features were engineered using the most important drivers combining ROC, SHAP and CM. Ex: Cross antenna mean, sum, or difference of RSSI to capture spatial shape characteristics of signals. (Fig. 13)

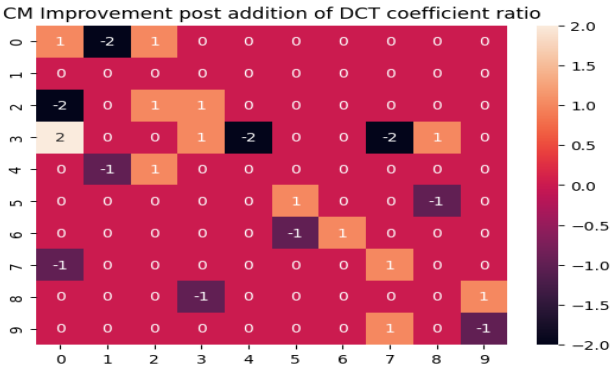

Fig. 7. Class-wise improvements in CM due to feature addition.

V. Experiments

1) Validation strategy: K-Fold vs Group K-Fold: K-Fold Cross-Validation randomly splits the dataset into K folds, using each fold once for validation. Although it provides a reliable estimate of average performance, it may lead to data leakage in CSI datasets due to temporal correlations between packets. Hence, Group K-Fold Cross-Validation was employed, ensuring that samples belonging to the same group were kept together and never split across training and validation folds. This prevents the model from over-relying on time-specific structures.

2) Seed Stability Analysis: To assess robustness, validation was repeated with different random seeds. Consistent Macro F1-scores indicate stable learning of CSI patterns rather than dependence on specific splits. XGBoost showed stable cross seed performance in comparison to RF and SVM, supporting its selection.

3) DCT v PCA: The amplitude dimensionality reduction was explored under both DCT ($\geq$ 99% energy) (Fig 8) and Principal Component Analysis ($\geq$ 90% Variance). Based on cross-validation, the usage of DCT provided better model performance, and hence DCT was retained. (Fig 9)

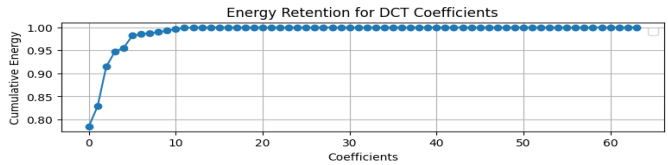

Fig. 8. Energy Retention by Cosine Coefficients

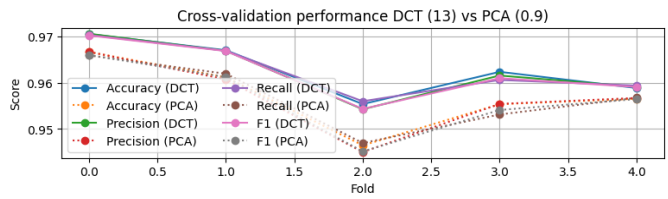

Fig. 9. CV DCT v/s PCA.

4) Hyperparameter tuning via grid search: XGBoost hyperparameters were tuned using grid search under group K-Fold cross-validation (Tab. III). Although tuning results in moderate F1-score improvement, the calibration curves before and after tuning remain largely similar (Fig. 11). This suggests that tuning primarily improves class separability. The predicted probabilities were reasonably calibrated under the logloss objective, and further gains were driven by improved discrimination between overlapping locations. (Fig. 10)

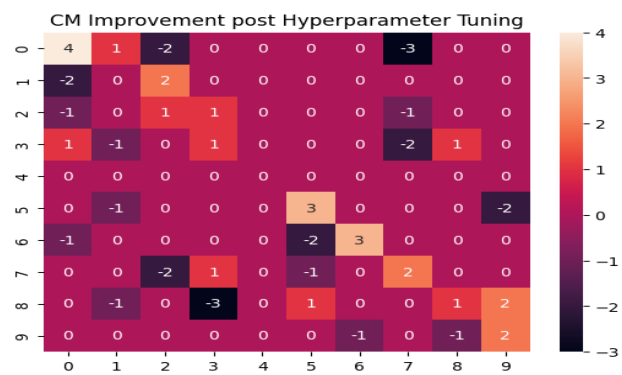

Fig. 10. Class-wise improvements in CM post tuning.

5) Regularization to Reduce Overfitting: Given the high-dimensionality of feature space, controlling overfitting and improving generalization is imperative. For XGBoost, Regularization was controlled mainly through grid search tuning of `reg_lambda` (L2 regularization), `max_depth`, and `min_child_weight` parameters.

6) Correlation with Timestamp: The A/B testing with and without Time-related features, such as delta or cumulative

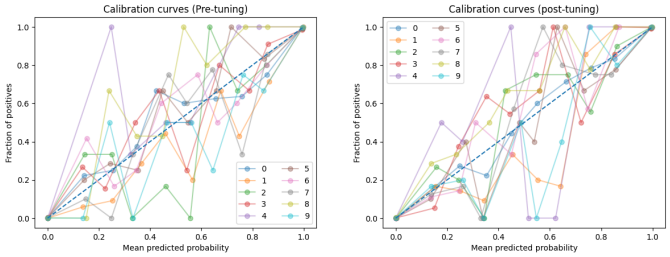

Fig. 11. Calibration Curves before and after Tuning.

time-period, revealed a significant difference in F1-score (Fig. 12). This stipulates CSI data may have been collected in a sequential order of positions or timestamp might be acting as a weak proxy for environmental conditions. While useful, timestamp features may introduce bias and should be handled carefully hence group-K fold offers a valid use case.

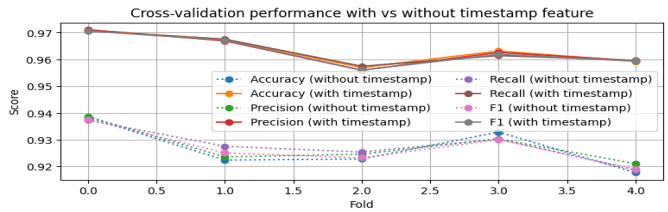

Fig. 12. Impact of Time-Related Features.

VI. Results

Among all evaluated models, XGBoost (XGB) achieves the best performance for CSI-based localization. The optimized hyperparameters for the final XGB model are reported in Tab. III.

SHAP identified top 6 features driving model decisions are presented in Fig. 13, dominated by timestamp descriptors, RSSI and DCT-transformed amplitude coefficients. The inverse-relation with position distance of RSSI can be clearly detected from SHAP summary plot.

TABLE III
XGBoost hyperparameter configuration.

Hyperparameter	Value
Objective	multi:softmax
Number of classes	$\lvert\mathcal{Y}\rvert$
Number of estimators	600
Maximum tree depth	5
Learning rate	0.05
Subsample ratio	0.8
Column subsample ratio	0.8
L2 regularization (λ)	1
Class weight	balanced
Minimum child samples	5
Evaluation metric	mlogloss
Random seed	42

Final Cross-validation scores under Group-k Fold are presented in Fig. 12 using the best-performing hyperparameters.

In general, combining RSSI with DCT-denoised amplitude features effectively distinguishes positions. XGBoost provides stable performance along with interpretable feature importance and class-level errors.

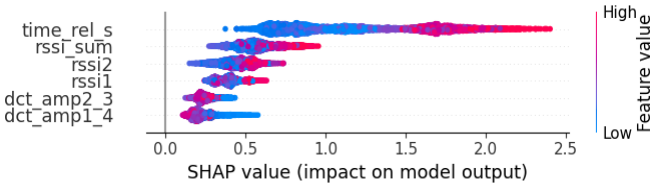

Fig. 13. Feature Type with Maximum Impact (SHAP summary plot).

TABLE IV
Δ F1 from Ablation experiments.

Features engineered	Δ F1
Raw IQ to derived Amplitude	+0.0996
DCT Amplitude [components=13]	+0.0170
RSSI descriptors	+0.0021
Max PSD on amplitude	+0.0019
Phase slope & delay	+0.0012
SHAP: Dynamic Feature Selection	+0.0009
Derived Angle of Arrival	+0.0007

VII. Conclusion

In this paper, we studied indoor localization as a multi-class classification problem using Wi-Fi Channel State Information (CSI). By transforming raw CSI measurements into structured fingerprints, spatial information related to distance and angle can be captured using standard Wi-Fi signals without additional hardware. Feature engineering played a key role in improving performance. RSSI captured distance-related patterns, while DCT-denoised amplitude features stabilized CSI fingerprints, reducing noise and high dimensionality. The Timestamp-related analysis further showed that temporal information can influence localization performance.

Among the evaluated classifiers, XGBoost with engineered CSI features achieves good performance. However, a strong dependence on timestamp-related features suggests that future work should be validated under session-independent and environment-independent conditions. Overall, the proposed pipeline shows that engineered CSI features combined with ensemble-based learning can be used to achieve stable performance in a controlled experimental set-up.

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